

Homework 4

CENG 113, SP 2026

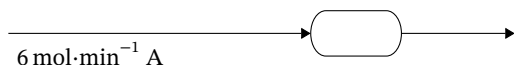
Dr. Drews

Academic Integrity: For all problems **show your work** to receive credit. By submitting your assignment you're confirming that your submission is a product of your own effort.

Box all answers!

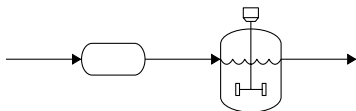
Problem 1

Consider again the irreversible reaction $A \longrightarrow 2B$ from Problem 2 on Homework 3. The fresh feed is pure A at $6 \text{ mol} \cdot \text{min}^{-1}$ and the rate constant is 4 min^{-1} .



(1.A) Find the conversion of A if the residence time in a PFR is 10 s.

(1.B) Calculate the outlet molar flow rate of A if a CSTR is added directly after the PFR, as shown below. The residence time in both reactors is 10 s and the rate constant is the same for both reactors.

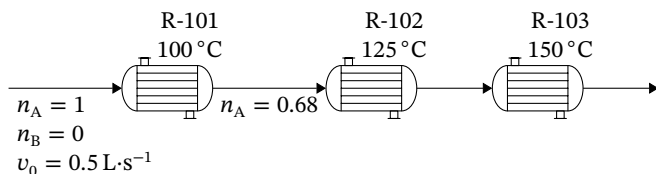


(1.C) Repeat Part B if the order of the reactors is swapped such that the CSTR is first and the PFR is second.

(1.D) Find the outlet molar flow rate if the two reactors are operated in parallel with the feed (both n_{A0} and v_0) are divided equally.

Problem 2

The liquid-phase, elementary reaction $A \longrightarrow B$ ($E_a = 21 \text{ kJ} \cdot \text{mol}^{-1}$) is to take place in a series of isothermal shell-and-tube PFRs, each 20 L, operating at different temperatures (flows are $\text{mol} \cdot \text{s}^{-1}$):



(2.A) Determine the pre-exponential factor A .

(2.B) Determine the outlet molar flow rate of A from R-103.

(2.C) Suppose the flow of steam provided to the heating jacket on R-102 falls a little bit, resulting in R-102's temperature falling to 110°C . To what temperature should R-103 be set to achieve the same outlet molar flow rate of A that you found in Part B? (ANS: $150 < T(^{\circ}\text{C}) < 175$)

Problem 3

The elementary, liquid-phase isomerization of normal butane to isobutane has a rate constant of 31 h^{-1} at 360 K and an activation energy of $65.7 \text{ kJ} \cdot \text{mol}^{-1}$. The feed is pure normal butane at $145 \text{ kmol} \cdot \text{h}^{-1}$, $9.3 \text{ kmol} \cdot \text{m}^{-3}$, and 330 K .

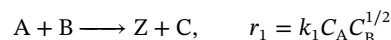
(3.A) Create a 1-by-3 subplot figure (see MATLAB's subplot command, as well as Example 7.5) with molar flow rate profiles on the left, reactor temperature profile in the center, and conversion profile of normal butane on the right plot for a 3 m^3 adiabatic reactor. The x -axis on all plots should be reactor volume in m^3 . (ANS: $X_A \approx 0.7$)

(3.B) Suppose the reactor volume is to be reduced to 2 m^3 with a diameter of 15 cm , and a utility jacket is added. Assuming a phase change fluid as a utility, at what temperature should this utility be provided to ensure approximately the same conversion found in Part A? (recall that "phase change" means the utility temperature will be constant) (ANS: $T_u \approx 340 \text{ K}$ or 360 K)

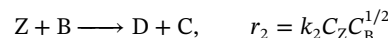
Additional information: $K_C = 3$ at 60°C , $\Delta H^\circ = -6.9 \text{ kJ} \cdot \text{mol}^{-1}$, $C_p = 141 \text{ J} \cdot \text{mol}^{-1} \cdot ^\circ\text{C}^{-1}$ (both compounds).

Problem 4

The desired product Z is formed according to the reaction



but the undesired side reaction



can also take place. The feed to an isothermal, isobaric, gas-phase PFR operating at 833 K and 35 atm contains $66.7 \text{ mol}\%$ B and $33.3 \text{ mol}\%$ A with a total volumetric flow rate of $13.5 \text{ m}^3 \cdot \text{h}^{-1}$. Create a plot of the outlet **concentration** of Z as a function of reactor volume. What volume should be used to maximize the concentration of Z? (ANS: $2.5 \text{ m}^3 < V < 3.0 \text{ m}^3$)

Additional information:

$$k_1 = 13.8 \left(\frac{\text{m}^3}{\text{kmol}} \right)^{0.5} \text{ h}^{-1}$$

$$k_2 = 7.5 \left(\frac{\text{m}^3}{\text{kmol}} \right)^{0.5} \text{ h}^{-1}$$